MIL-T-25624F
22 June 1978
SUPERSEDING
MIL-T-25624E
28 June 1973

MILITARY SPECIFICATION

TRANSMITTER, PRESSURE TRU-80/A, OIL, ENGINE-MOUNTED, 0-100 PSI

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of remote-indicating 320° movement, synchro style, differential, engine-mounted oil-pressure transmitter, designated TRU-80/A.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-763	Steel Bars, Shapes, and Forgings-Corrosion Resisting
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation-Packaging, Methods of
MIL-E-5400	Electronic Equipment, Airborne, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to:HQ AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6620

Source: https://assist.dla.mil -- Downloaded: 202640-20T06:50Z
Check the source to verify that this is the current version before use.

MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-I-7057	Indicators, Synchro, Aircraft, General Specification for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-S-20708	Synchros, General Specification for
MIL-B-27497	Bearings, Jewel, Sapphire or Ruby, Synthetic

STANDARDS

MILITARY

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods
MS33649	Bosses, Fluid Connection - Internal Straight Thread
MS33678	Connector, Receptacle, Electric, Integral Mounting

Source: https://assist.dla.mil -- Downloaded: 20261O-20T06:50Z
Check the source to verify that this is the current version before use.

HANDBOOKS

MILITARY

MIL-HDBK-275	Guide for Selection of Lubrications, Fluids, and Compounds for use in Flight Vehicles and Components

PUBLICATIONS

MILITARY

SD-6	Provisions Governing Qualification

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. Transmitters furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3). Qualification must be accomplished within the provisions and requirements of SD-6. That document requires that tests shall be monitored by a government representative designated by the preparing activity or its authorized agent. The test procedure shall be in accordance with this specification. All tests required by this specification shall be successfully accomplished and documented. Certification of test results is required by the witnessing government representative.

3.2 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials. Reclaimed materials shall be utilized to the maximum extent possible within the quality limits required by this document. Materials used in the manufacture of transmitters shall be suitable for the purpose and shall conform to applicable Government specifications. Materials conforming to contractor's specifications may be used provided the specifications are approved by the Government and contain provisions for adequate tests. The use of the contractor's specification will not constitute waiver of Government inspection.

Source: https://assist.dla.mil -- Downloaded: 202640-20T06:50Z
Check the source to verify that this is the currentversion before use.

3.3.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they are used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.3.2 Metals. Metals shall be corrosion-resistant or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.3.3 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.4 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the transmitter, except where magnetic materials are essential.

3.3.5 Corrosive fumes. Materials used in the construction of the transmitter shall not produce corrosive or deleterious fumes under the test conditions specified herein.

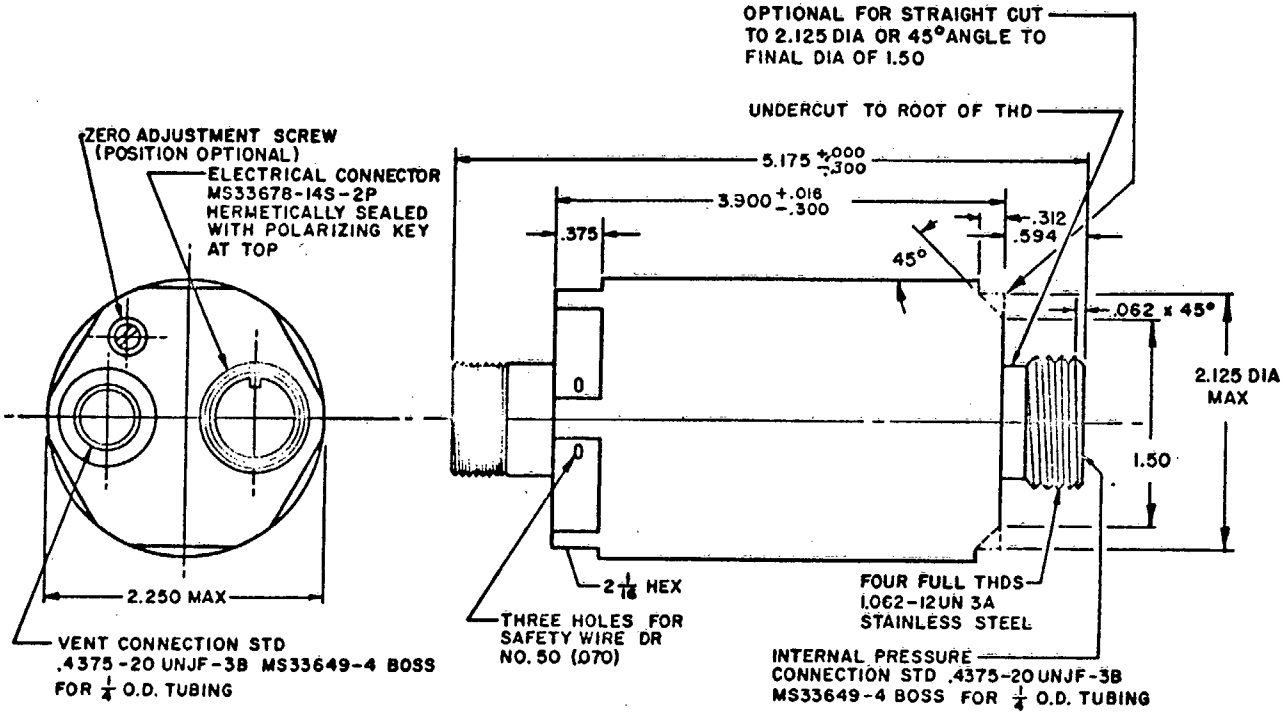
3.3.6 Protective treatment. When materials are used in the construction of the transmitter that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Design and construction.

3.4.1 Design. The transmitter shall be designed to operate with indicators conforming to MIL-I-7057.

3.4.1.1 Outline dimensions. The outline dimensions of the transmitter shall conform to figure 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:50Z
Check the source to verify that this is the current version before use.



DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: ANGLES ± 0° 15', DECIMALS ± .016

FIGURE 1. Transmitter Case and Connectors

Source: https://assist.dla.mil -- Downloaded: 202640-20T06:50Z
Check the source to verify that this is the current version before use.

3.4.1.2 Adjustments. The transmitter shall be provided with a means for adjusting or correcting indications which shall be simple to manipulate and such that adjustments can be accomplished with tools ordinarily available. The means of adjustment shall be accessible from the exterior of the case and shall provide a minimum of ±7 angular degrees of synchro rotor adjustment from electrical zero.

3.4.1.3 Mechanisms. Transmitter mechanisms shall be comprised of units distinct from their cases. The units shall be sufficiently rugged so that small distortions of the cases will not affect the accuracy of the indications.

3.4.1.4 Adjustment locks. All calibration adjustments shall be provided with securing devices to lock the adjusting mechanism in place to prevent the adjustment from drifting from its original setting due to environmental conditions specified herein. Provisions for locking shall be made on the calibration adjustment so that it cannot be readily moved.

3.4.1.5 Reliability. The transmitter shall have a specified meantime between failures (MTBF) of 2,000 hours at a 90 percent confidence level.

3.4.1.6 Lubricant. The contractor shall select a lubricant or lubricants in accordance with MIL-HDBK-275 which will be suitable for use on gears, joints, bearings, and pinions of the transmitter. Prior to selecting the lubricant or lubricants, the contractor shall make an investigation relative to the lubricant most suitable for the application with due consideration being given to the environmental conditions under which the transmitter must operate. The type of lubricant and the reason for its selection shall be specified by the contractor and approved by the procuring activity.

3.4.1.7 Synchro transmitter. Each transmitter shall be capable of operating either one or two indicators, whose synchros conform to the electrical characteristics of MIL-S-20708/5, without change in indication of more than one degree of dial arc.

3.4.1.7.1 Synchro bearings. The precision ball bearings used in the synchro shall be corrosion-resistant alloy steel (stainless steel) conforming to QQ-S-763, class 440C, and shall be made to ABEC tolerance 5P.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:50Z
Check the source to verify that this is the current version before use.

3.4.1.7.2 Brushless synchro. The brush and slipring method shall not be used in the synchro transmitter to transport excitation power to the synchro rotor. The excitation power shall be transported by means of a continuous electrical circuit which shall permit the rotor to turn readily through an arc of 340° minimum. Stops shall be provided to prevent overwinding. A means shall be provided for readily setting the synchro at electrical zero. The synchro transmitter design shall be such that the synchro electrical zero position can be set and locked in position so securely that it will remain in that position for the service life of the transmitter or until reset.

3.4.1.8 Overpressure stop. An overpressure stop shall be provided to restrain the prime mover. The stop shall permit the transmitter rotor to rotate 12° ±3° beyond the 100-psi position. The stop shall not be damaged by application of 500 psi overpressure.

3.4.1.9 Gears, pivots, joints, bearings. Before finalizing the design of the pressure transmitter mechanism, the contractor shall investigate and determine the best known types and the proper processing of interfacing materials for use in the gears, pivots, joints, and bearings in the train of moving parts of the transmitter. These interface materials shall be selected for their ability to resist wear, corrosion, and friction when interfacing with one another. The interfacing materials shall not be dissimilar as described in MIL-STD-889. After selection of the material types and processing procedures, the contractor shall submit a description of each interfacing material and process proposed for the train of moving parts (gears, pivots, joints, bearings) and present his reason for selection of each type to the procuring activity for approval.

3.4.1.9.1 Large-area interface surface. The gears, pivots, joints, and bearings shall have substantially large interface surfaces for the purpose of minimizing the effects of wear upon the calibration and accuracy of the transmitter. Single point type interfaces, such as attained by pivots made up of two perpendicular cylinders will not be acceptable, while interface surfaces made up of two parallel cylinders or curved surfaces will be acceptable if sufficient interface area is provided to minimize the effects of wear. The contractor shall furnish explicit details in his engineering report concerning this design factor and shall show how his proposed design will minimize wear.

Source: https://assist.dla.mil -- Downloaded: 202640-20T06:50Z
Check the source to verify that this is the currentversion before use.

3.4.1.10 Attitude. The transmitter shall be so designed that it will operate as specified herein in any position.

3.4.1.11 Differential pressure. The differential pressure transmitter shall have two pressure connections (internal pressure connection and vent connection) as shown in figure 1. The transmitter shall be so designed and constructed that it will be actuated by the difference between the pressure applied to the internal pressure connection and the pressure applied to the vent connection, and the synchro signal output shall represent internal pressure minus vent pressure. The angular position of the rotor of the synchro relative to the stator shall be essentially a linear function of internal pressure minus vent pressure.

3.4.2 Construction. The transmitter shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipment, storage, installation, and service. Pivots, bearings, and gears shall not bind nor shake and shall be as nearly frictionless as possible.

3.4.2.1 Mounting torque. The transmitter shall be of such rigid construction that it will withstand a torque of 720 inch-pounds applied to the 2-1/16-inch hex nut (see figure 1) without distorting the case or affecting the accuracy.

3.5 Performance. The transmitter shall be designed and calibrated to operate continuously and produce its greatest accuracy at a normal operating temperature of 177°C. The transmitter shall withstand and operate while being subjected to the following conditions:

a. Temperatures: Operating temperatures ranging from -54° to +177°C and exposure to temperatures ranging from -62° to +232°C

b. Vibration: Vibration of 40g from 169 to 2,000 Hz

c. Altitude-temperature: Absolute pressure equivalent to an 85,000-foot altitude at -54°C

d. Salt fog: Exposure to salt sea atmosphere for 48 hours

e. Overpressure: An overpressure of 200 psi for a 10-minute period

f. Humidity: Exposure to humidity for 240 hours (non-hermetically sealed transmitters only).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:50Z
Check the source to verify that this is the current version before use.

3.5.1 Dielectric strength. The transmitter shall withstand 200V at a commercial frequency applied between any pin of the electrical connector, not connected to the case, and the case for a minimum period of 5 seconds.

3.5.2 Voltage and frequency variation. The transmitter shall withstand voltage and frequency variations from 22 to 30V and 320 to 480 Hz or any combinations thereof.

3.5.3 Electromagnetic interference. The transmitter shall meet the electromagnetic interference requirements of MIL-STD-461 as specified for class A1 equipment.

3.5.4 Damping. When the transmitter is subjected to an applied oil pressure sufficient to produce an indication of 100 psi and the pressure is quickly released, the time required for the pressure to drop from 100 to 10 psi shall be 1.0 to 4.0 seconds when the transmitter is at -54°C.

3.5.5 Indicator operation. The transmitter shall be capable of operating either one or two indicators without a change in output equivalent to more than 1° of arc.

3.5.6 Vent boss and connector. The vent boss and connector shall withstand an applied force of 25 foot-pounds bending moment with no evidence of damage to welds or soldering, distortion of the base plate, and without the calibration shifting out of tolerance.

3.5.7 Scale error. Scale error shall not exceed the tolerances specified in table I.

3.5.8 Friction and position error. Friction and position error shall not exceed the tolerances specified in table II.

3.5.9 Effect of lead length. When the lead length (No. 20 AWG stranded copper wire) between the transmitter and indicator synchro stators is changed from 5 to 300 feet, the indication shall not change more than 1° of arc.

Source: https://assist.dla.mil -- Downloaded: 202640-20T06:50Z
Check the source to verify that this is the currentversion before use.

TABLE I. Scale Error Tolerances (± PSI)

Test Pressure PSI	Room Temp, High Temp, Normal Operating Temp (177°C)	AFTER: Humidity, over pressure, High Temp Exposure & During Room Temp, During & After Reliability	DURING: Low Temp, Low Temp. Exposure, Vibration at High Temperature, Sensors Matching
0	2.0	3.0	4.0
20	2.0	3.0	4.0
30	1.5	2.5	3.0
40	1.5	2.0	2.5
45	1.5	2.0	2.5
50	1.5	2.0	2.5
60	1.5	2.0	2.5
70	1.5	2.5	3.0
80	2.0	3.0	4.0
100	2.0	3.0	4.0

TABLE II. Friction and Position Error

Tests	Tolerances (± PSI)
Friction:	
High Temperature	2.0
High Temperature Exposure	2.0
Low Temperature	2.0
Low Temperature Exposure	2.0
Position Error at 50 PSI:	0.5
Maximum pointer oscillation	1.0
Maximum pointer variation	1.0

3.5.10 Leakage

3.5.10.1 Hermetically sealed case. When a pressure of approximately 0 psia is applied to the exterior of the transmitter case, the leak rate shall not exceed 0.10 micron cubic foot per hour when 10-percent helium is used in the filling medium and shall not exceed 1.0 micron cubic foot per hour when 100 percent helium is used as the filling medium.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:50Z
Check the source to verify that this is the current version before use.

3.5.10.2 Nonhermetically sealed case. There shall be no leakage when the transmitter is submerged in a partially filled container of water which has been evacuated to an air pressure of 1.0 psi.

3.5.11 Differential Pressure. The differential pressure transmitter shall be capable of receiving input pressures varying from 0 to 100 PSI applied to the internal pressure connection in excess of the vent pressure and input pressures varying from 0 PSI to 40 PSI applied to the vent connection in any pressure combinations as long as the input pressure exceeds the vent pressure. Scale error shall not exceed the tolerances specified in Table I for "Sensors Matching" at room temperature.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part number and changes thereto.

3.7 Case. The case shall conform to figure 1 and shall provide either a hermetically sealed or gasket sealed enclosure for the internal mechanism of the transmitter. Sealing of the case shall be so accomplished that the seal will not be dependent upon materials which would be affected by the action of any atmosphere to which the transmitter may be subjected.

3.7.1 Filling medium. When a hermetically sealed case is used, the filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling medium shall be either a mixture of 88 to 92 percent nitrogen and the remainder helium or 100 percent helium. The absolute pressure of the filling medium in the case shall be 12 to 14.7 psi.

3.7.1.1 The transmitter shall not contain silicone oil.

3.8 Electronic components. Electronic components and the application thereof shall be in accordance with MIL-E-5400. Nonstandard parts shall not be used except with prior approval of the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 202640-20T06:50Z
Check the source to verify that this is the currentversion before use.

3.9 Electrical connector. An MS33678-14S-2P hermetically sealed electrical connector shall be provided.

3.10 Electrical characteristics. The transmitter synchro shall meet the requirements of 3.10.1 through 3.10.3 with a normal excitation of 26V 400-Hz single-phase power applied to the primary (rotor) from a source having electrical characteristics conforming to MIL-STD-704 as specified for category B equipment.

3.10.1 Open secondary. The current drawn by the primary of the self-synchronous motor shall not exceed 240 ma with secondaries (stator) open circuited.

3.10.2 Power input. Power consumption of the transmitter synchro shall not exceed 6.24 va.

3.10.3 Secondary voltage. The maximum open-circuit voltage measured between any two secondary leads shall be 11.8 ±0.3V. The variation between the three maximum secondary voltages on any one synchro shall not exceed 0.2V.

3.11 Wiring.

3.11.1 External. The electrical receptacle shall be so wired that the 26-V, 400-Hz power supply will be connected to terminals A and B. External wiring between the transmitter and indicator shall be as shown on figure 2.

3.11.2 Internal. Internal wiring of the transmitter shall be color coded and shall be connected to the receptacle pins as shown on figure 3. When gears are used to drive the transmitter counterclockwise for increasing values of the parameter being measured, terminals S1 and S3 shall be reversed internally to cancel the effects of the gear reversal to assure clockwise rotation of the indicator synchro shaft and pointer with increasing values.

3.12 Jewel bearings. When used, synthetic sapphire or ruby jewel bearings shall be in accordance with MIL-B-27497.

3.13 Finishes and protective coatings.

3.13.1 Aluminum alloy parts. Aluminum alloy parts shall be coated with an anodic film conforming to MIL-A-8625 except that small holes, straight threads, case inserts, and parts in hermetically sealed enclosures need not be anodized.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:50Z
Check the source to verify that this is the current version before use.

3.13.1.1 Aluminum alloys that do not anodize satisfactorily shall be coated with a chemical film conforming to MIL-C-5541.

3.13.1.2 When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.14 Weight. The weight of the transmitter shall not exceed 1.75 pounds.

3.15 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.16 Identification of product. Equipment, assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

3.16.1 Nameplate. A nameplate shall be permanently attached to the rear of the case or the required information shall be permanently etched on the side of the case where paint is not needed as a protective coating. Where applicable, the information marked thereon shall conform to MIL-STD-130. The nameplate shall not be glued or cemented to the case nor shall metals decals be used. Etching shall be permanently inscribed so that it will withstand all normal environments for the life of the transmitter.

3.17 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9.

3.17.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimenisons and tolerances affect the interchangeability, operation, or performance of the transmitter, they shall be held or limited accordingly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 202640-20T06:50Z
Check the source to verify that this is the currentversion before use.

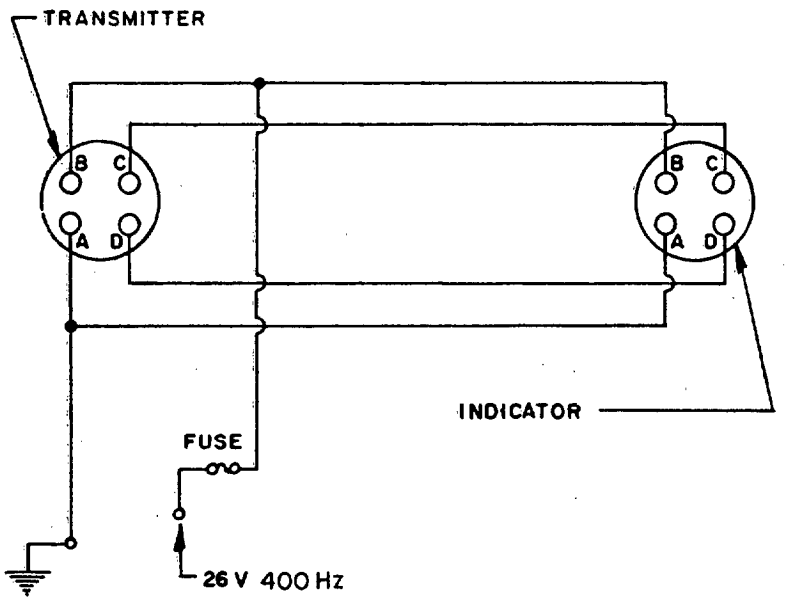


FIGURE 2. Wiring Diagram - External, Single Indicator

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:50Z
Check the source to verify that this is the current version before use.

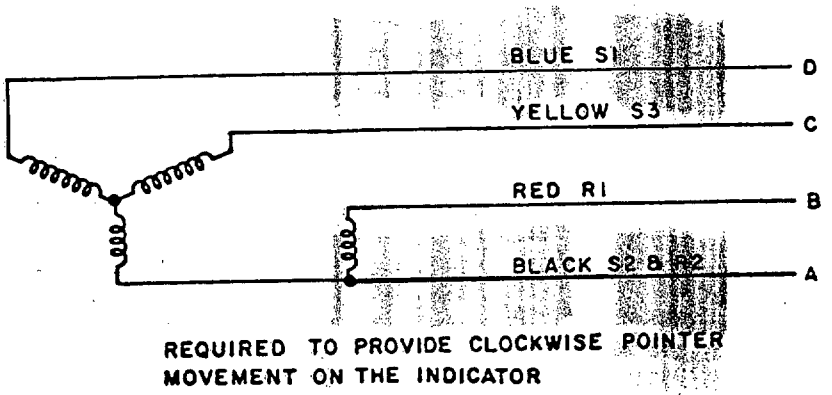


FIGURE 3. Internal Wiring - Transmitter

Source: https://assist.dla.mil -- Downloaded: 20260-20T06:50Z
Check the source to verify that this is the current version before use.

4.2 Classification of tests. The inspection and testing of transmitters shall be classified as follows:

a. Qualification testing . . . . . See 4.4

b. Quality conformance tests . . . . . See 4.5.

4.3 Test conditions.

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, the test shall be conducted at an atmospheric pressure of approximately 29.92 inches Hg and at an ambient temperature of $177^{\circ}{}^{+5}_{-0}{}^{\circ}C$. For tests conducted at room temperature, the temperature shall be $25^{\circ} \pm 5^{\circ}C$.

4.3.2 Test voltage. The transmitter shall be tested with 26V 400-Hz single-phase power. Unless otherwise specified, the voltage and frequency shall be maintained within ±1 percent of the nominal value.

4.3.3 Tapping. The transmitter shall be lightly tapped or vibrated before each scale error test reading.

4.3.4 Test standard. The transmitter shall be tested against an indicating standard having a maximum error of ±0.1 of a circular degree. The primary transmitting master shall be a precision deadweight tester which shall have a maximum inherent error of ±0.03 psi.

4.3.5 Adjustment. Synchro rotor adjustment shall not be used to bring scale errors within tolerances after tests are initiated except that when a failure is charged to a scale error shift during reliability testing, an adjustment may be made.

4.4 Qualification testing.

4.4.1 Test samples. The test samples consist of a minimum of 10 transmitters representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:50Z
Check the source to verify that this is the current version before use.

4.4.1.1 Disposition of samples upon completion of tests. Any sample tested may be delivered on contract, provided it meets the following:

a. It is representative of production units currently being accepted

b. It is in good-as-new condition or has been refurbished (see 6.4.1 and 6.4.2)

c. It is otherwise satisfactory.

4.4.2 Qualification tests. The qualification tests shall consist of all the tests specified under 4.6.

4.5 Quality conformance tests. Quality conformance tests shall consist of:

a. Individual tests . . . . . .See 4.5.1

b. Sampling tests . . . . . .See 4.5.2.

4.5.1 Individual tests. Each transmitter shall be subjected to the following tests as described under 4.6:

a. Examination of product . . . . . See 4.6.1

b. Scale error test . . . . . See 4.6.2

c. Friction error test . . . . . See 4.6.3

d. Position error test . . . . . See 4.6.4

e. Dielectric strength test . . . . . See 4.6.5

f. Leakage tests . . . . . See 4.6.6.1 and 4.6.6.2

g. Electrical zero and verification of leads . . See 4.6.7

h. Sensors matching (20 PSI Vent Pressure) . . See 4.6.26.1.

Source: https://assist.dla.mil -- Downloaded: 202640-20T06:50Z
Check the source to verify that this is the currentversion before use.

4.5.2 Sampling tests. One transmitter shall be selected at random from each one hundred or less produced on the contract or order and subjected to the following tests as described under 4.6:

a. Individual tests . . . . . .See 4.5.1

b. Low temperature test . . . . . .See 4.6.8

c. High temperature test . . . . . .See 4.6.9

d. Vibration test . . . . . .See 4.6.10

e. Electrical characteristics . . . . . .See 4.6.11

f. Sensors matching . . . . . .See 4.6.26.

4.5.2.1 Rejection and retest. When one item selected from a production run fails to meet the specification, transmitters on hand or later produced shall not be accepted until the extent and cause of failure have been determined.

4.5.2.2 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of transmitters on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in transmitters already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods.

4.6.1 Examination of product. The transmitter shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, marking, dimensions, and weight.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:50Z
Check the source to verify that this is the current version before use.

4.6.2 Scale error test. The transmitter shall be tested for scale error at both normal operating and room temperatures at the test points specified in table I for both upscale and downscale readings. The test points on the upscale reading shall be measured by bringing the input reading up to, but not exceeding, the specified test point. The difference in the scale error after vibration from the input value shall not exceed the tolerances specified in table I. The downscale error shall be measured in the same manner, except that the input reading value shall be brought down to, but not below, the specified test point.

4.6.3 Friction error test. The friction error test shall be conducted at both normal operating and room temperatures and may be performed concurrently with the scale error test. The difference in readings before vibrating the transmitter and after vibration has been applied shall be the friction error which shall not exceed the tolerances specified in table II.

4.6.4 Position error test. The position error test may be conducted in conjunction with the test specified in 4.6.2, except at room temperature only. The transmitter shall be energized to obtain a reading of approximately one-half scale. The change in indication produced by rotating the transmitter from the normal operating position to a position 90° clockwise and counterclockwise (longitudinal axis vertical) shall be noted. The maximum change in indication shall not exceed the tolerance specified for position error in table II.

4.6.5 Dielectric strength test. This test shall be conducted at room temperature. A potential of 200V rms at a commercial frequency shall be applied between each pin of the electrical connector, not connected to the case, and the case for a minimum period of 5 seconds. There shall be no insulation breakdown.

4.6.6 Leakage tests.

4.6.6.1 Hermetically sealed case. The hermetically sealed transmitter shall be tested for leaks at room temperature by means of a mass-spectrometer-type helium leak detector. The detected leak rate for the 10-percent helium and 90-percent nitrogen filling medium shall not exceed 0.1 micron cubic foot per hour and for the 100-percent-helium filling medium shall not exceed 1.0 micron cubic foot per hour at an absolute pressure differential of 12 to 14.7 psi.

Source: https://assist.dla.mil -- Downloaded: 202640-20T06:50Z
Check the source to verify that this is the currentversion before use.

4.6.6.2 Nonhermetically sealed case. The nonhermetically sealed transmitter shall be immersed in water in a container which shall be filled to not more than 70 percent of its capacity. The water level shall be 2 to 4 inches above the nearest part of the transmitter housing. The container shall first be evacuated to 0.50 psi air pressure and held at this pressure for 1 minute or until air bubbles cease to be given off by the liquid, whichever is greater. The pressure shall then be raised to 1.0 psi and held at this pressure for a minimum period of 5 minutes. Any bubbles resulting from air or fluid leakage from within the case shall be cause for rejection. Bubbles that are the result of entrapped air on the various exterior parts of the case shall not be considered as leaks.

4.6.7 Electrical zero and verification of terminal leads. This test shall be conducted at room temperature. The transmitter shall be wired in accordance with figure 3. Pressure shall be applied at the pressure port of the transmitter to obtain the following conditions to determine electrical zero and to verify the terminal leads:

a. The voltage between S1 and S3 shall be a minimum, shall be in phase with rotor voltage E (R2R1), and shall remain approximately in phase as the rotor is turned counterclockwise through an angle not exceeding 180° from electrical zero

b. The voltage between R1 and S3 shall be near the low null and shall reach the low null when the rotor is turned clockwise approximately 30° form electrical zero

c. Clockwise rotation of the rotor not exceeding 30° from electrical zero shall cause an increase in voltage between S3 and S2 and a decrease in voltage between S2 and S1.

4.6.7.1 Electrical zero adjustment. With pressure applied, the transmitter shall be subjected to, and shall meet, the test specified in 4.6.7 during which the zero adjustment shall be operated to the extreme positions. The required minimum adjustment shall be $\pm$7 angular degrees from the electrical zero position.

4.6.8 Low temperature test. The transmitter shall be subjected to a low temperature test in accordance with MIL-STD-810, method 502, at -54 $\pm$2°C for a period of 4 hours after which it shall be subjected to the scale error and friction error tests while still at that temperature. The errors shall not exceed the tolerances specified in tables I and II, respectively, for low temperature.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:50Z
Check the source to verify that this is the current version before use.

4.6.9 High temperature test. The transmitter shall be placed in a chamber and subjected to a temperature of $177^{\circ}\ ^{+5^{\circ}}_{-0^{\circ}}$ C for a 4-hour period. Throughout this test, the transmitter shall be electrically connected to an appropriate indicator with power applied. At the end of the 4-hour period and with the transmitter still at the specified temperature, the scale error and friction error tests shall be conducted. The scale error and friction error shall not exceed the tolerances specified in tables I and II.

4.6.10 Vibration test. The transmitter shall be subjected to a vibration test in accordance with MIL-STD-810, method 514, equipment category (b), along a curve parallel to curve L of figure 514.1-2 at a vibration level of 40g at frequencies ranging from 169 to 2,000 Hz. During this test, the transmitter shall meet the variation and oscillation requirements of table II. The vibration level from 5 to 169 Hz shall follow curve L. This test shall be conducted with the transmitter subjected to a temperature of $177^{\circ}\ ^{+5^{\circ}}_{-0^{\circ}}$ C. At the end of this test, the transmitter shall meet the test specified in 4.6.6.1 or 4.6.6.2, as applicable.

4.6.11 Electrical characteristics. The transmitter shall be tested to determine compliance with the requirements specified herein with respect to primary current, power input, secondary voltage, and operation of either one or two indicators with a change in indication of not more than 1° of arc.

4.6.12 Low temperature exposure test. The transmitter shall be subjected to a low temperature exposure test in accordance with MIL-STD-810, method 502. The transmitter shall be exposed to a temperature of $-62^{\circ} \pm 2^{\circ}$C for a minimum period of 48 hours after which it shall be operated throughout its entire range. It shall exhibit no erratic characteristics during operation at the $-62^{\circ} \pm 2^{\circ}$C temperature. The temperature shall then be raised to $-54^{\circ} \pm 2^{\circ}$C and the transmitter maintained at this temperature for a minimum of 24 hours. At the end of the 24-hour period, the transmitter shall be subjected to the scale error and friction error tests. The scale error and friction error shall not exceed the tolerances specified in tables I and II for low temperature exposure.

Source: https://assist.dla.mil -- Downloaded: 202640-20T06:50Z
Check the source to verify that this is the currentversion before use.

4.6.13 High temperature exposure test. The transmitter shall be placed in a chamber and subjected to a temperature of 230° ±5°C for a period of 80 hours. At the end of this period and while still at the specified temperature, the transmitter shall be subjected to the scale error test. The transmitter shall then be returned to room temperature for 4 hours after which it shall again be subjected to the scale error test. The scale errors shall not exceed the tolerances specified in table I for during and after high temperature exposure.

4.6.14 Altitude-temperature test. The transmitter shall be subjected to an absolute pressure equivalent to an altitude of 85,000 feet for a period of 1 hour at a temperature of -54° ±2°C. While still at this pressure and temperature, the transmitter shall be subjected to the scale error test. The scale error shall not exceed the tolerances specified in table I for low temperature.

4.6.15 Electromagnetic interference. The transmitter shall be subjected to the electromagnetic interference tests specified in MIL-STD-461 up to 400 MHz for class ID equipment. The methods used shall be as specified in MIL-STD-462.

4.6.16 Humidity test (nonhermetically sealed case). The humidity test shall be conducted in accordance with MIL-STD-810, method 507, procedure I. Any evidence of corrosion, other deterioration, or moisture penetration shall be cause for rejection. At the completion of this test, the error shall not exceed the tolerances specified in table I for scale error at room temperature and at 177°C.

4.6.17 Salt-fog test. The salt-fog test shall be conducted in accordance with MIL-STD-810, method 509, at room temperature for a period of 48 hours. The transmitter shall be examined and there shall be no evidence of external corrosion or deterioration that would affect subsequent operation. The transmitter shall then meet the test specified in 4.6.5. Before the dielectric strength test is conducted, the external surface of the transmitter may be washed with distilled water and air dried.

4.6.18 Voltage and frequency variations. The transmitter shall be subjected to the scale error and friction error tests during which the voltage and frequency shall be varied from 22 to 30V and 320 to 480 Hz, or any combinations within these

Source: https://assist.dla.mil -- Downloaded: 20261O-20T06:50Z
Check the source to verify that this is the current version before use.

specified ranges. The scale and friction errors shall not exceed those specified in tables I and II, respectively. This test shall be conducted at room temperature.

4.6.19 Effect of lead length. The lead length between the transmitter and indicator synchro stators shall be changed from 5 to 300 feet of No. 20 AWG stranded copper wire. The indication shall not change more than 1° of arc. This test shall be conducted at room temperature.

4.6.20 Vent boss and connector strength test. The vent boss and connector mounting shall be sufficiently rigid to withstand a force of 25 foot-pounds bending moment applied in any direction perpendicular to the longitudinal axis of the transmitter without the calibration shifting out of tolerance, distortion of the base plate, or evidence of damage to the vent boss, connector welds, or soldering. This test shall be conducted with the transmitter secured by the engine mount connection or by the adjacent hex nut only. The transmitter shall then be subjected to and shall meet the sealing test. This test shall be conducted at room temperature.

4.6.21 Overpressure tests.

4.6.21.1 Overpressure (200 psi). This test shall be conducted at room temperature. An overpressure of 200 psi shall be applied to the transmitter for 10 minutes. Not less than 1 hour following this test, the transmitter shall be subjected to the scale error test on increasing pressure only. The scale error shall not exceed the tolerances specified in table I for after overpressure. Application of the 200-psi overpressure shall position the pointer of the indicating standard off scale approximately midway between the 100- and 0-psi graduations as required for the overpressure stop specified in 3.4.1.8.

4.6.21.2 Overpressure (500 psi). This test shall be conducted at room temperature with the case immersed in water and only one sample transmitter shall be used. The case shall be punctured or the seal shall be broken and a pressure of 500 psi shall be applied at the input pressure port. Leakage of the internal mechanism through the hole in the case shall be cause for rejection. The sample transmitter need not be checked for accuracy following this test. The transmitter shall then be disassembled and inspected for compliance with 3.4.1.7.1, 3.4.1.7.2, 3.4.1.9, and 3.4.1.9.1.

Source: https://assist.dla.mil -- Downloaded: 202640-20T06:50Z
Check the source to verify that this is the current version before use.

4.6.22 Mounting torque test. The mounting torque test shall be conducted at room temperature. One transmitter shall be mounted on a suitable mounting pad and a force of 720 inch-pounds shall be applied radially to the 2-1/16-inch hex nut and in a direction perpendicular to the longitudinal axis of the transmitter for a period of 2 minutes. At the conclusion of this test, the transmitter shall be subjected to the scale error test. Distortion of the case or failure to meet the accuracy requirements shall be cause for rejection.

4.6.23 Damping test. A pressure sufficient to produce full-scale deflection shall be applied to the transmitter. A quick-release valve shall be used to release the pressure, and the time for the indication to change from 100 to 10 psi shall be 1.0 second to 4.0 seconds. This test shall be conducted with oil, such as grade 1005 or equivalent, applied at the pressure port at a temperature of -54° $\pm 2^{o}C$. Neither restrictors nor a capillary line shall be used to connect the pressure input to the pressure element. This test shall be conducted at -54° $\pm 2^{o}C$.

4.6.24 Helium test. When the transmitter is hermetically sealed, the case shall be punctured or the filling tube cut and the case tested by means of a mass-spectrometer-type helium leak detector. Failure to detect helium leakage shall be cause for rejection.

4.6.25 Reliability tests.

4.6.25.1 Test program. Prior to starting the reliability test, the contractor shall prepare a detailed reliability test program in accordance with MIL-STD-781 and submit it to the procuring activity for formal approval.

4.6.25.2 Qualification test procedure. Each sample transmitter shall be subjected to a minimum of 1,000 hours of operation consisting of an oil-line pressure oscillation at a frequency of 60±3 cpm from 35±1 to 50±1 psi. Temperature cycling shall be in accordance with MIL-STD-781, test level J, except the high temperature shall be $177^{o}C$. Once in approximately each 16 hours the pressure cycling shall be stopped for a period of not longer than 1 minute during which the pressure shall be reduced to 0 psi before resuming the cycling. The scale error at room temperatuer test shall be conducted at least once every

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:50Z
Check the source to verify that this is the current version before use.

200 hours. Throughout the pressure cycling, for 10 minutes of each hour the transmitter shall be subjected to sinusoidal vibration cycling in accordance with MIL-STD-810, method 514.2, along a curve parallel to curve L (figure 514.2-2) at 30g and at frequencies from 5 to 2,000 Hz. After completion of this test, the accuracy of each sample transmitter shall be within the tolerances specified in table I for during and after reliability. During each accuracy check, a failure shall be counted for each sample transmitter that does not meet the specified tolerances. Accept-reject criteria shall be in accordance with MIL-STD-781, test plan I. No preventive maintenance shall be performed. Reliability test records shall also be in accordance with MIL-STD-781.

4.6.25.3 Production sampling (for Army or Navy contracts only). Sample transmitters shall be subjected to a production reliability test in accordance with MIL-STD-781,test level F, except the upper temperature level shall be 145°C and the vibration frequency cycle shall be from 5 to 2,000 Hz and return during each 10-minute vibration period in accordance with MIL-STD-810, curve T, figure 514.1-5. Accept-reject criteria shall be in accordance with test plan II. This test shall be conducted on at least one sample from each lot. Transmitters produced during 1 month shall constitute a lot. The number of samples shall be as specified by the procuring activity. Testing shall be started with the first month's production and the samples shall be tested until an accept or reject decision is reached. Unless otherwise specified in the contract or order, no transmitters shall be shipped until an accept decision is reached for each lot.

4.6.26 Sensors matching. The transmitter adjustment shall be adjusted, and locked, for operation with the test indicator and the adjustment shall not be further disturbed for the balance of the "Sensor Matching" test. For each of the following conditions, the transmitter shall be tested and meet the scale error (sensors matching), at room temperature, at the test points specified in table I for both up-scale and down-scale readings. The test points on the up-scale reading shall be measured by bringing the input reading up to, but not exceeding, the specified test point. The down-scale error shall be measured in the same manner, except that the input reading value shall be brought down to, but not below, the specified test point.

Source: https://assist.dla.mil -- Downloaded: 202640-20T06:50Z
Check the source to verify that this is the currentversion before use.

4.6.26.1 20 PSI vent pressure. A 20 PSI pressure shall be applied to the vent connection of the transmitter. Pressures shall be applied to the internal pressure connection which shall exceed that applied to the vent connection by each of the test points listed in table I.

4.6.26.2 40 PSI vent pressure. A 40 PSI pressure shall be applied to the vent connection of the transmitter. Pressures shall be applied to the internal pressure connection which shall exceed that applied to the vent connection by each of the test points listed in table I.

4.7 Inspection and testing of preparation for delivery. Preparation for delivery shall be tested in accordance with MIL-P-116 to insure compliance with section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be in accordance with 5.1.1 or 5.1.2, as specified (see 6.2).

5.1.1 Level A (reference definition in MIL-STD-794). Transmitters shall be preserved and packaged in accordance with MIL-P-116, method 1C-1, one each in a container conforming to PPP-B-636, weather-resistant class. Cushioning shall be of a type, density, configuration, and thickness that will insure that shock transmission will not exceed the peak shock g-value specified for transmitters when completed packs are submitted to the rough-handling drop tests of MIL-P-116.

5.1.2 Level C (reference definition in MIL-STD-794). Transmitters shall be packaged in accordance with 5.1.1, except method III of MIL-P-116 shall apply in lieu of method 1C-1.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Transmitters preserved and packaged as specified in 5.1.1 shall be packed in weather-resistant shipping containers conforming to PPP-B-636. Shipments that exceed the weight limitations of the specification shall be packed in overseas-type containers conforming to PPP-B-601. The gross weight of each PPP-B-601 container shall not exceed 200 pounds. Where

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:50Z
Check the source to verify that this is the current version before use.

practicable, exterior shipping containers shall be of uniform shape and size and of minimum cube and tare. Closure and strapping shall be in accordance with the appendix of PPP-B-636 or PPP-B-601, as applicable. Waterproofing shall be in accordance with MIL-STD-1186.

5.2.2 Level B. Transmitters preserved and packaged as specified in 5.1.1 shall be packed in domestic shipping containers conforming to PPP-B-636. Where practicable, exterior shipping containers shall be of uniform shape and size and of minimum cube and tare. The gross weight of each shipping container shall not exceed the weight limitations of the specification. Closure and strapping shall be in accordance with the appendix of PPP-B-636.

5.2.3 Level C. Packages that require overpacking for acceptance by the carrier shall be placed in exterior-type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall comply with Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

5.3 Physical protection. Cushioning, blocking, bracing, and waterproofing shall be in accordance with MIL-STD-1186. Waterproofing requirements for cushioning materials and containers shall be waived for domestic shipments. The drop tests of MIL-STD-1186 shall be waived when the item is preserved, packaged, and packed for immediate use or when the drop tests of MIL-P-116 are applicable.

5.4 Marking. In addition to any other marking required by the order or contract (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129. The shipment marking nomenclature shall be:

TRANSMITTER, PRESSURE TRU-80/A, OIL, ENGINE-MOUNTED, 0-100 PSI

5.4.1 Reinspection date marking. The reinspection date marking shall be as specified by the procuring activity (see 6.2).

6. NOTES

Source: https://assist.dla.mil -- Downloaded: 202640-20T06:50Z
Check the source to verify that this is the currentversion before use.

6.1 Intended use. The TRU-80/A transmitter is intended for use in transmitting, by electrical signal, the oil pressure in the lubricating system of aircraft engines to an indicator or indicators.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Information, if any, to be submitted with the qualification samples

c. Levels of packaging and packing required, special shipment marking, and reinspection date marking (see section 5).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Hq AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016. Information pertaining to qualification of products to this specification may be obtained from that activity.

6.4 Definitions. For the purpose of this specification, the following definitions will apply:

6.4.1 Good as new. Good as new means the transmitter has been operated less than 10 percent of the specified MTBF operation.

6.4.2 Refurbished. Refurbished means that the transmitter has been completely overhauled with all component parts meeting current part standards, and the transmitter has been subjected to and met all the requirements for a new transmitter.

6.5 Marginal indicia. Asterisks are not used in this revision to identify the changes with respect to the previous issue due to the extensiveness of the changes.

6.6 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:50Z
Check the source to verify that this is the current version before use.

Custodians:
Navy - AS
Air Force - 99

Preparing Activity:
Air Force - 99

Project Number:
6620-0313

☆U.S. GOVERNMENT PRINTING OFFICE: 1978 — 703-122/3357

Source: https://assist.dla.mil -- Downloaded: 202640-20T06:50Z
Check the source to verify that this is the current version before use.